

ECSE 2500

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April 6

Plan: Joint CDF/PMF/PDF of two RVs

→ Marginal CDF/PMF/PDF of one RV

→ Independence of two RVs

□ From joint PMF of two RVs to
Marginal PMF of each RV

Given joint PMF $P_{X,Y}(x,y)$, $\forall x \in S_X, y \in S_Y$, we want to
 $\uparrow$
all values RV X can take

marginal PMF $P_X(x) = \sum_{y_i \in S_Y} P_{X,Y}(x, y_i)$

likewise, $P_Y(y) = \sum_{x_i \in S_X} P_{X,Y}(x_i, y)$

This definition means that we can look at the PMF
of X and Y on their own, by ignoring the other variable.

Example

One random experiment generates two RVs

Flip coin n times. Define

$X(s) = \#$ of heads over n coin flipping

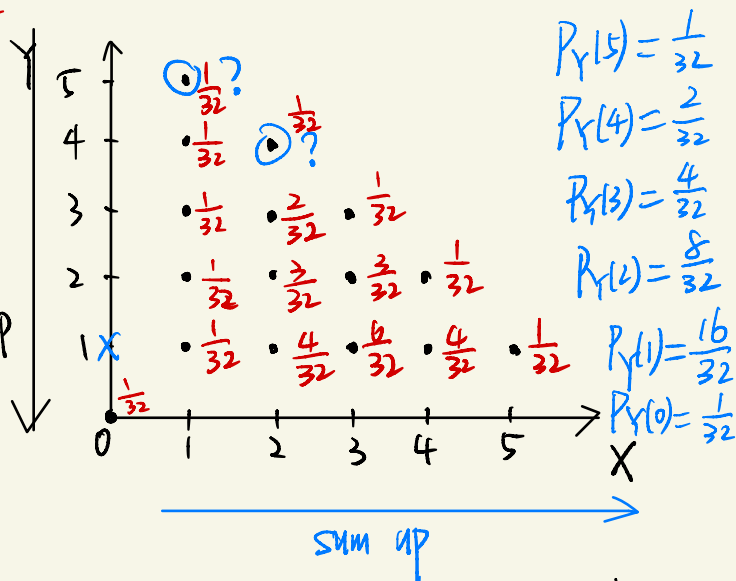
$Y(s) =$ position of the first head

Copy from last lecture

Joint PMF

$P_{XY}(x,y)$

Sum up



$$P_X(0) = \frac{1}{32} \quad P_X(1) = \frac{5}{32} \quad P_X(2) = \frac{10}{32} \quad ; \dots \quad P_X(5) = \frac{1}{32}$$

Note: $P_X(x)$ is a Binomial distribution, and
marginal PMF $P_Y(y)$ is a Geometric distribution.
This confirms that we can obtain the information
of each RV from joint PMF.

In the same way, this is a notion of marginal CDF

$$\begin{aligned} F_X(x) &= P(X \leq x) \\ &= F_{X,Y}(x, \infty) = P(X \leq x, Y < \infty) \end{aligned}$$

$$F_Y(y) = F_{X,Y}(\infty, y) = P(X < \infty, Y \leq y)$$

Note: Just knowing the marginal PMF/CDF is not enough to obtain the joint PMF/CDF.

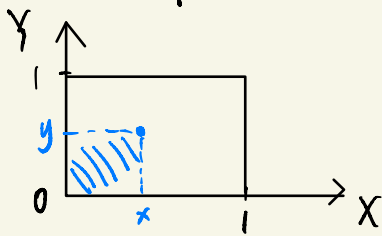
□ Joint CDF/PDF of two continuous RVs

For discrete RVs, joint PMF $F_{X,Y}(x,y)$ is a

"staircase" function, but for continuous RVs,

$F_{X,Y}(x,y)$ is a smooth increasing function.

Ex. Two Uniform random variables on unit square



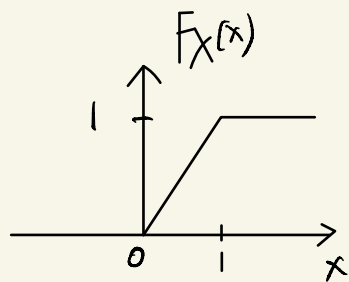
$$F_{X,Y}(x,y) = \begin{cases} 0, & \text{if } x \text{ or } y < 0 \\ xy, & \text{if } x, y \in [0,1] \\ x, & \text{if } x \in [0,1], y > 1 \\ y, & \text{if } x > 1, y \in [0,1] \\ 1, & \text{if } x, y > 1 \end{cases}$$

For continuous RVs, we can also define the notion of a marginal CDF. e.g.,

$$F_X(x) = P(X \leq x) = P(X \leq x, Y < \infty) \\ = F_{X,Y}(x, \infty)$$

We can apply it to the above joint CDF.

$$F_X(x) = \begin{cases} 0 & , \text{ if } x < 0 \\ x & , \text{ if } x \in [0, 1] \\ 1 & , \text{ if } x > 1 \end{cases}$$



This is the CDF of **one** uniform RV!

→ PDF For a continuous RV, we can obtain PDF by differentiating CDF.

$$F_X(x) = \int_{-\infty}^x f_X(z) dz$$

$$f_X(x) = \frac{d}{dx} F_X(x)$$

For the uniform RVs case,

$$f_X(x) = \begin{cases} 0, & \text{otherwise} \\ 1, & x \in [0, 1] \end{cases}$$

For the general case,

$$F_{X,Y}(x,y) = \int_{-\infty}^x \int_{-\infty}^y f_{X,Y}(u,v) du dv$$

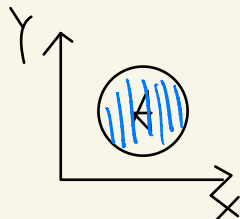
$$f_{X,Y}(x,y) = \frac{d^2}{dx dy} F_{X,Y}(x,y)$$

The joint PDF also has the property like PDF for single RV, e.g.,

$$\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f_{X,Y}(x,y) dx dy = 1$$

Probability of $(X,Y) \in A$ is

$$P(A) = \iint_A f_{X,Y}(x,y) dx dy$$

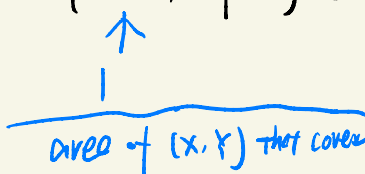


In particular, for a "Product-type" event,

$$P(a \leq X \leq b, c \leq Y \leq d) = \int_a^b \int_c^d f_{X,Y}(x,y) dx dy$$

Example: Jointly uniform RVs on $[0,1] \times [0,1]$
 (see the example in the beginning of class)

$$f_{X,Y}(x,y) = \frac{\partial^2 F_{X,Y}(x,y)}{\partial x \partial y} = \begin{cases} 0, & \text{otherwise} \\ 1, & \text{if } x,y \in [0,1] \end{cases}$$



If we want to calculate joint PDF of
 two RVs uniformly distributed over $[-2,5] \times [1,3]$

$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{14}, & \text{if } x \in [-2,5], y \in [1,3] \\ 0, & \text{otherwise} \end{cases}$$

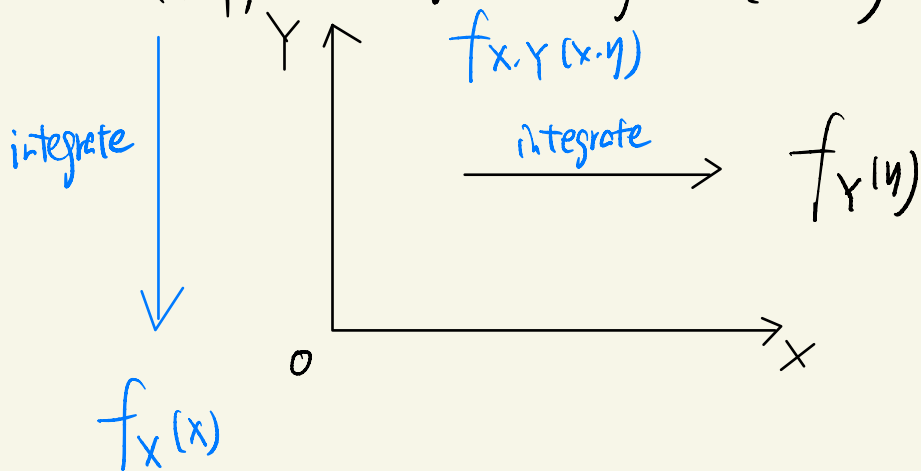
This is because integration of joint PDF $\rightarrow 1$.

□ From joint PDF to marginal PDF

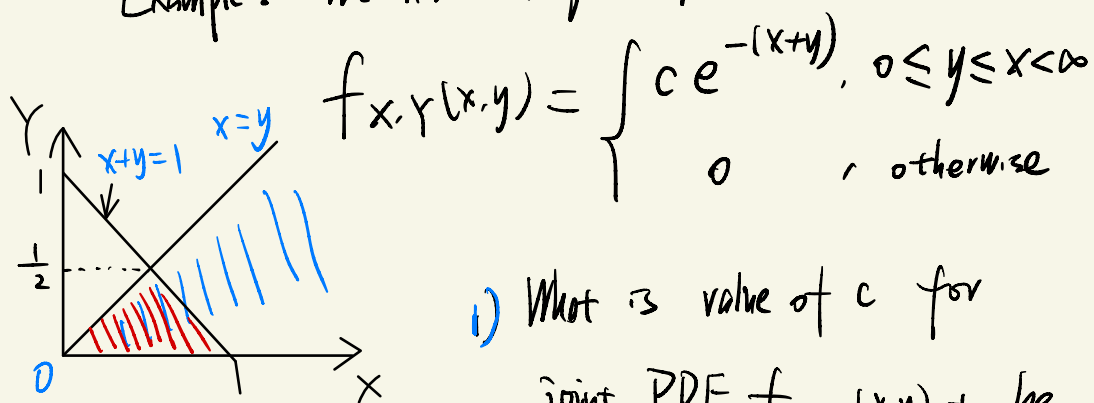
Marginal of X

$$\begin{aligned} f_X(x) &= \frac{d}{dx} F_X(x) = \frac{d}{dx} \underline{F_{X,Y}(x, \infty)} \\ &= \frac{d}{dx} \int_{-\infty}^x \int_{-\infty}^{+\infty} \underline{f_{X,Y}(u,v) du dv} \\ &= \int_{-\infty}^{+\infty} \underline{f_{X,Y}(x,y) dy} \quad \text{integrate over } y \end{aligned}$$

This is, to get the marginal in X , we integrate along Y (columns); to get the marginal in Y , we integrate along X (rows)



Example: We have a joint PDF



i) What is value of c for joint PDF $f_{X,Y}(x,y)$ to be a valid PDF?

i) $f_{X,Y}(x,y) \geq 0$

ii) $\iint f_{X,Y}(x,y) dx dy = 1$

$$\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} c e^{-(x+y)} dx dy$$

integration by parts

$$= c \cdot \frac{1}{2} = 1 \Leftrightarrow c = 2$$

2) What is the marginal PDF $f_X(x)$?

$$\begin{aligned} f_X(x) &= \int_{-\infty}^{+\infty} f_{X,Y}(x,y) dy \\ &= \int_{-\infty}^{+\infty} 2 e^{-(x+y)} dy \\ &= 2 e^{-x} \cdot (-e^{-y}) \Big|_{y=0}^{y=x} \\ &= \begin{cases} 2 e^{-x} (1 - e^{-x}), & \text{if } x \geq 0 \\ 0, & \text{if } x < 0 \end{cases} \end{aligned}$$

3) What is $P(X+Y \leq 1)$?

$$\begin{aligned} P(X+Y \leq 1) &= \int_0^{\frac{1}{2}} \left(\int_y^{1-y} 2 e^{-x} dx \right) e^{-y} dy \\ &= \int_0^{\frac{1}{2}} 2 e^{-y} (-e^{-x}) \Big|_y^{1-y} dy \\ &= \int_0^{\frac{1}{2}} 2 e^{-y} (e^{-y} - e^{-(1-y)}) dy = 1 - 2e^{-1} \end{aligned}$$

Important example of joint PPF of two continuous RVs

One RV $\frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

Joint Gaussian RVs

$$f_{X,Y}(x,y) = \frac{1}{2\pi |\Sigma|^{\frac{1}{2}}} e^{-\frac{1}{2} \left(\begin{bmatrix} x \\ y \end{bmatrix} - \underline{\mu} \right)^T \Sigma^{-1} \left(\begin{bmatrix} x \\ y \end{bmatrix} - \underline{\mu} \right)}$$

mean vector $\underline{\mu} = \begin{bmatrix} \mu_x \\ \mu_y \end{bmatrix} \in \mathbb{R}^2$; $\Sigma = \begin{bmatrix} \sigma_x^2 & \rho\sigma_x\sigma_y \\ \rho\sigma_x\sigma_y & \sigma_y^2 \end{bmatrix} \in \mathbb{R}^{2 \times 2}$

$\uparrow$
covariance matrix

$\uparrow$
critical parameter $\rho \in [-1, 1]$
captures correlation between X and Y .